Employee Attrition Analysis Report

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About the data. The analysis used the IBM HR Analytics Employee Attrition dataset, a public file of 1,470 employee records. IBM data scientists created it, and it does not come from a real company. The findings show how the method works. They should not be read as a description of any actual employer. The report tends to answer three business questions: where do we lose people, what it costs and what to do about it.

Summary

Out of the 1,470 employees in the data, 237 left, an attrition rate of 16.1%. If replacing a leaver costs half a year's salary, which is the planning assumption used throughout, those departures cost about USD 6.8 million.

Table 1: Summary Statistics

N	Number of leavers	Attrition rate	Estimated replacement cost
1,470	237	16.1%	US$6.8m

The losses are concentrated. Employees who work overtime left at 30.5%, compared with 10.4% for those who do not. They make up 28% of the workforce but 54% of the leavers. The sharpest case is Level 1 staff on overtime: 156 people, of whom 82 left (52.6%). The second hotspot is new joiners. Staff with a year of service or less left at 34.9%, nearly double the 18.4% seen at two to three years.

We recommend a pilot of two measures: overtime relief for Level 1 roles, and structured support through the first year of employment. On placeholder costs of USD 274,000 a year, the scenario model puts the net benefit at about USD 1.4 million, and the programme breaks even if it closes only 8% of the gap. Those figures rest on assumptions, and the pilot exists to test them before anything is rolled out more widely.

Where people are leaving

Sales has the highest departmental rate at 20.6%, followed by Human Resources at 19.0% and Research & Development at 13.8%. Research & Development lost the most people (133) because it is the largest department. Its replacement bill of about USD 3.3 million is almost the same as Sales, which lost 92 people but on higher salaries (average USD 70,900 against USD 49,300 for leavers in Research & Development).

Roles show a wider spread. Two in five Sales Representatives left (39.8%). Laboratory Technicians lost the most people of any role, 62 in all (23.9%). Human Resources staff left at 23.1%, though on a base of only 52. Research Directors (2.5%) and Managers (4.9%) rarely leave.

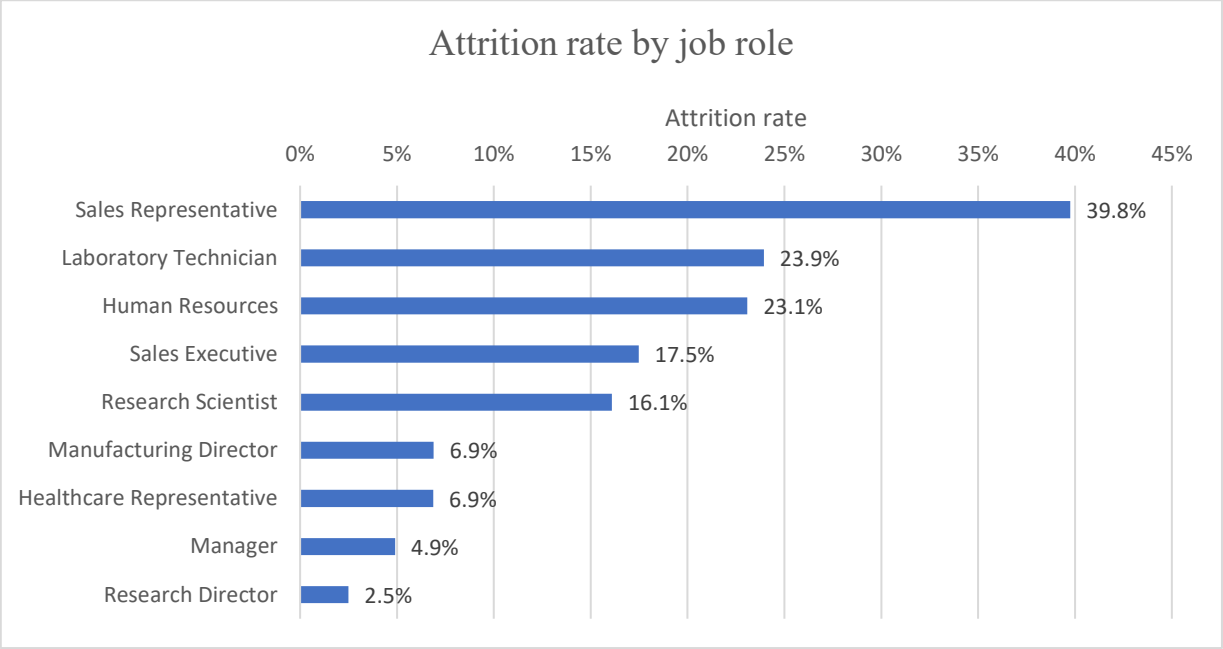


Figure 1: Attrition rate by job role

Overtime

Overtime is the strongest single pattern in the data. It shows up at every job level but is far larger at the bottom. At Level 1, 52.6% of employees working overtime left, against 15.8% of those who did not. At Level 2 the figures are 17.8% and 6.7%. Level 1 overtime workers are about 11% of the workforce and produced 35% of all leavers.

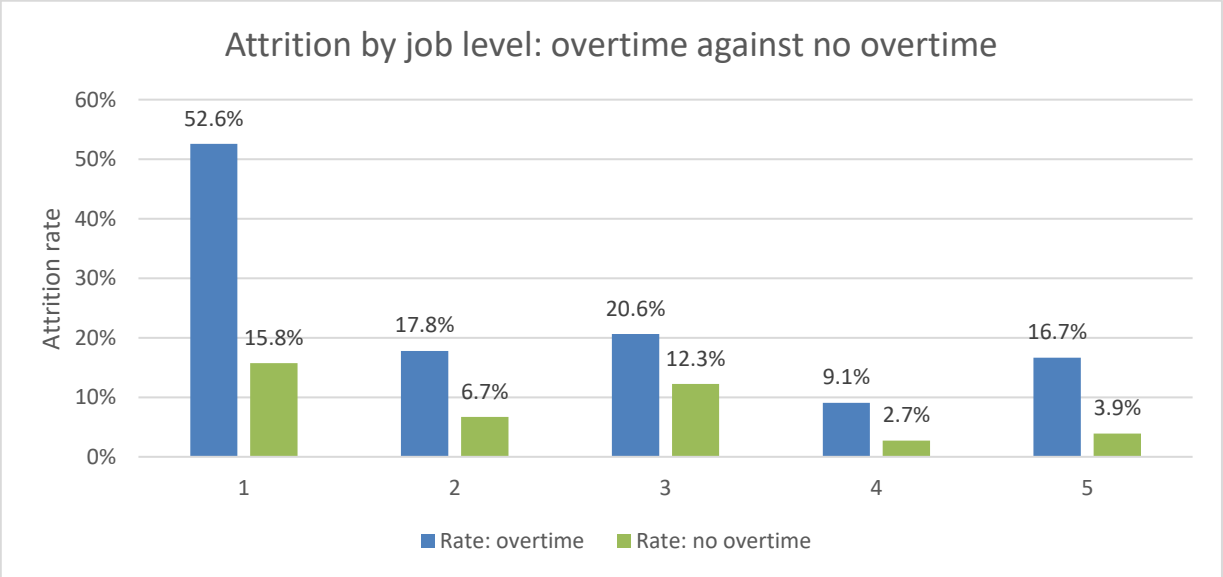


Figure 2: Attrition by job level

The riskiest segment we found is Level 1 staff on overtime with a year of service or less: 31 of 44 left (70.5%). Frequent business travellers also left at a high rate, 24.9% against 8.0% for staff who do not travel.

Length of service

Attrition is highest early and falls in every later band. Staff with a year of service or less left at 34.9%. By two to three years the rate is 18.4%, and beyond ten years it is 8.1%. Almost a third of all leavers (75 of 237) had been with the company for a year or less.

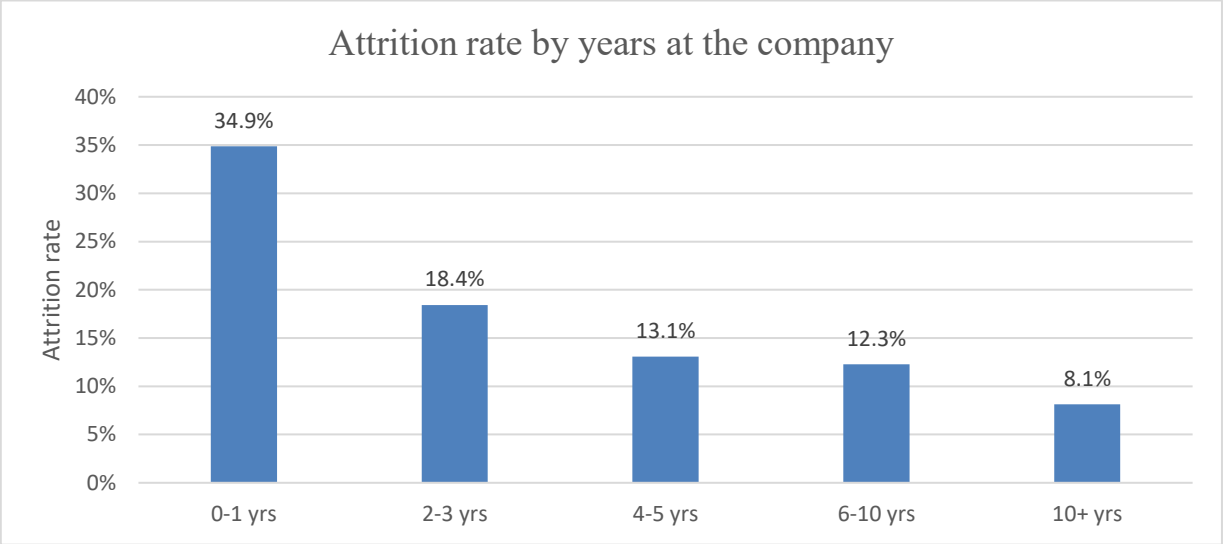


Figure 3: Attrition rate by years of service

Pay and other factors

Pay matters, but less than workload. Employees paid below 85% of the average for their job level left at 20.8%. Those paid 115% or more left at 13.1%. In four roles (Research Scientist, Sales Representative, Human Resources and Laboratory Technician) leavers earned between 12% and 17% less than colleagues who stayed. In senior roles the pattern reverses. Leavers among Research Directors, Healthcare Representatives and Sales Executives earned more than stayers, so pay does not explain those exits.

Two other gaps are worth noting. Employees with no stock options left at 24.4%, compared with 9.4% at option level 1. Employees who rated their work-life balance as low left at 31.3%, although only 80 people gave that rating.

Predicting who is at risk

To move from group patterns to individual follow-up, we built a model that scores each employee's likelihood of leaving. It uses 17 fields, including overtime, travel, job level, department, pay, tenure, promotion history, satisfaction ratings and commute distance. We tested it by cross-

validation, so every score came from a model that had never seen that employee's record. The area under the curve was 0.83, where 0.5 is a coin toss and 1.0 is perfect.

In practice, flagging the 20% of staff with the highest scores would catch 62% of the people who left. Half of those flagged did leave, three times the base rate of 16%. The High band left at 45.8% and the Low band at 3.9%.

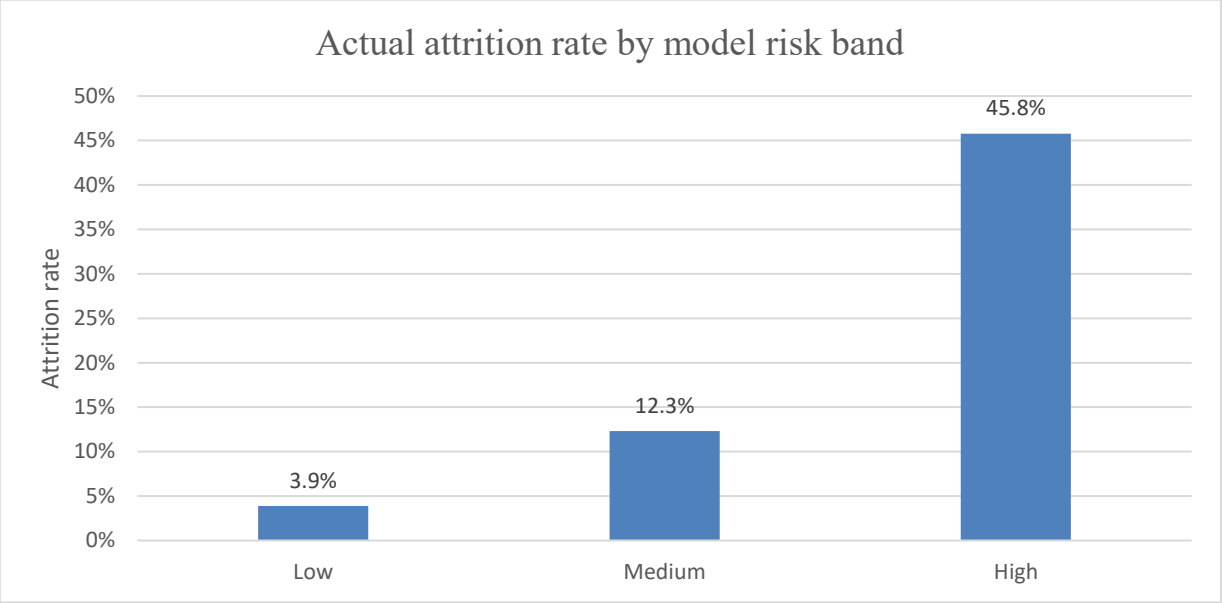


Figure 4: Actual attrition rate by model risk band

Overtime is the strongest factor in the model. With everything else held equal, an employee on overtime has about 4.5 times the odds of leaving. Marital status and gender were left out on purpose. They predict well in this dataset but are not fair grounds for HR action. The scores are for prioritising retention conversations with line managers. They should not be used for selection, promotion or disciplinary decisions.

Business case

The scenario assumes a pilot that closes half of the gap between each hotspot and its comparison group. Overtime staff move halfway toward the rate for staff without overtime, and first-year staff move halfway toward the rate for years two and three.

Table 2: Group Ccomparison

Group	Headcount	Current rate	Target rate	Leavers avoided	Programme cost (USD)
Overtime staff	416	30.5%	20.5%	41.8	166,400
First-year staff	215	34.9%	26.7%	17.7	107,500
Combined				**59.5**	**273,900**

Table 3: Net f cost

Annual figures	**USD**
Gross saving (leavers avoided x replacement cost)	2,126,508
Less 20% overlap between the two groups	(425,302)
Net saving	1,701,206
Programme cost	273,900
Net benefit	**1,427,306**

Net of cost, the return is about USD 5.20 for every dollar spent, with payback in roughly two months. The result holds under most alternatives we tried. If replacement costs only a quarter of salary and the programme closes just 10% of the gaps, it loses about USD 104,000 a year. The other eleven combinations tested (replacement cost from 25% to 75% of salary, gaps closed from 10% to 75%) all show a gain.

These are not forecasts. The programme costs are placeholders, and the replacement-cost share is a planning figure that the company should swap for its own. The data is one snapshot, so it shows that overtime and early tenure go with higher attrition. It does not prove that cutting overtime will cut attrition by the amount assumed.

Recommendations

1. **Pilot overtime relief for Level 1 roles.** Start with the 156 Level 1 employees on overtime, where the problem is largest. Caps on monthly overtime, rotation of extra shifts and time off in lieu are the obvious levers to test.
2. **Support the first year.** Give every new joiner a named buddy, hold check-ins at 30, 90 and 180 days, and have a structured exit conversation with anyone who leaves in year one.
3. **Review pay in the hardest-hit roles.** Look first at staff paid below 85% of their level average in Sales Representative, Laboratory Technician and Research Scientist roles, where leavers earned noticeably less than stayers.
4. **Use the watchlist for retention conversations, and measure.** Share the top-risk list with line managers within the limits set out in section 6. Record results quarterly in the Benefits Tracker and revisit the assumptions after two quarters of actual figures.

Limits of this review

The data is a single snapshot with no dates, so we cannot see trends or judge whether the year was typical. Attrition is a yes/no field and does not separate resignations from dismissals or retirements.

Overtime is also yes/no, with no hours. The dataset is synthetic, and with 1,470 records the rates for small groups, such as Human Resources, Level 5 or Sales Representatives, can swing a lot from one year to the next.

Appendix: how the analysis was built

Stage	Tool	What it did
Clean and validate	Python (pandas)	Nine integrity checks, all passed. Dropped three columns that held one value for every row. Added tenure, age, promotion and commute bands.
Aggregate	SQL (SQLite)	Twelve queries on attrition by department, role, overtime, tenure and pay position, using joins, CTEs, CASE logic and window functions.
Model	Python (scikit-learn)	Logistic regression with 5-fold cross-validation. Scores written back to the database and joined in SQL to produce the watchlist.
Visualise and cost	Excel	Dashboard built on COUNTIFS and AVERAGEIFS formulas, scenario model with sensitivity grid, and a quarterly benefits tracker.